

Bhavya Dharmaraaj

Graduate Student
National Centre for Biological Sciences (NCBS)
Tata Institute of Fundamental Research (TIFR)
GKVK, Bellary Road,
Bangalore-560065, Karnataka, India

Mobile No: +91-9880653368
Email: bee.dee29@gmail.com /
bhavyad@ncbs.res.in
Twitter: @BeeBhavya

Education

2015 – present: Integrated Masters-PhD student at NCBS, Bangalore, India.

Advisor: Krushnamegh Kunte

2012 – 2015: B. Sc. in Microbiology, Chemistry, Zoology at St. Joseph's College, Bangalore, India.

Scholarships/awards received

- 1) **Best oral presentation:** Indian society of evolutionary biologists (ISEB) conference, 2024.
Talk Title: Sexual Selection and Functional Roles Influence Colouration but not the Amount of Variation in Butterfly Wing Colour Patterns
- 2) NCBS Bridging Postdoctoral Fellowship (February 2024 – August 2024)
- 3) **Best poster award:** Indian society of evolutionary biologists (ISEB) conference, 2023.
Poster title: Parallel Repeated Co-option of a Sexual Ornament for a Defensive Signal in Mimetic Evolution
- 4) **Best poster award** in evolutionary biology (NCBS annual talks, 2022)
Poster title: Parallel Repeated Co-option of a Sexual Ornament for a Defensive Signal in Mimetic Evolution
- 5) NCBS student fellowship (2015 – current)
- 6) B.Sc. Zoology gold medallist – 2016

Research experience

- 1) PhD student at the Biodiversity Lab, National Centre for Biological Sciences, Bangalore, India (August 2015 – Current). My thesis work involved understanding the evolution of wing colour diversity in butterflies. To that effect I used a number of techniques and analyses to characterize and understand wing colour variation.
 - a) I carried out chemical characterization of wing pigments from *Papilio* butterflies: I standardized the chemical extraction and have used Liquid Chromatography-Mass Spectrometry (LCMS) for identification of pigments from butterfly wings.
 - b) I characterized wing colours through UV and visible light photography, reflectance spectrophotometry and avian visual modelling.
 - c) I have experience with conducting behavioural experiments such as mate-choice and predation experiments.
 - d) Field experience: I have collected and processed data in multiple field sites within (Karnataka and West Bengal) and outside India (Ecuador).
 - e) I have experience coding in R where I wrote scripts for statistical analyses and data organisation.
 - f) I have extensively used Adobe Photoshop and Illustrator to prepare figures and schematics for my thesis and papers.
- 2) Research intern at Prof. Chandrabhas Narayana's lab in Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR), Bangalore India. (April-June 2014). I worked on two projects i) characterizing antimicrobial activity of silver-impregnated bacterial cellulose

and ii) the use of Surface Enhanced Raman Scattering (SERS) in detection of cytokines. I used chemical, microbial, and spectroscopic techniques for these projects.

Scientific Publications

- 1) **Dharmaraaj B.** and Kunte K. 2025. Natural and sexual selection and functional roles influence colouration but not the amount of variation in butterfly wing colour patterns. *BMC Ecol Evo* 25, 11. <https://doi.org/10.1186/s12862-024-02346-8>
- 2) **Dharmaraaj B.**, Venkatesan R., and Kunte K. 2024. Spectral variation and pigmentary basis of ornamental and mimetic wing colour patches of swallowtail butterflies. *Evol. J. Linn. Soc.*, doi: 10.1093/evolinnean/kzae018.
- 3) Outomuro, D, Zurek, D.B., Taylor L.A., Cronin T.W., **Dharmaraaj B.**, Kunte, K., Morehouse N. The evolution of colour vision across jumping spiders. *Integr. Comp. Biol.* **59**, E176 (2019) (**Conference proceedings**)

Other Appointments

- 1) I was part of an exchange program with Colorado State University (CSU), Fort Collins, for a period of 3 weeks in February 2018.
- 2) Member of the [Elife](#) Community Ambassadors program for the year 2022–2023.
- 3) Member of the *Proceedings B* Preprint Editorial team for the subfield of Evolutionary Biology (August 2023–Present)

Conferences/Workshops/Symposia

- 1) Delivered a talk at the ISEB-2024 conference.
Talk Title: Sexual Selection and Functional Roles Influence Colouration but not the Amount of Variation in Butterfly Wing Colour Patterns
- 2) Presented a poster at the ISEB-2023 conference.
Poster Title: Parallel Repeated Co-option of a Sexual Ornament for a Defensive Signal in Mimetic Evolution
- 3) Presented a talk at ESEB-2022 in the symposium: Predator cognition and the evolution of prey defence strategies.
Poster Title: Parallel Repeated Co-option of a Sexual Ornament for a Defensive Signal in Mimetic Evolution
- 4) Attended the ESEB-2021 online satellite symposium on “Predator cognition and evolution of prey defence strategies”.
- 5) Presented a virtual poster at the online Evolution-2021 conference.
Poster Title: Wing colour variation across butterfly species.
- 6) Presented a poster at ESEB-2019 held in Turku, Finland in the symposium, “Colour across the evolutionary spectrum: from production to perception”.
Poster Title: Effect of natural and sexual selection on wing colour variation in butterflies.
- 7) Volunteered and attended the Biology of Butterflies conference in 2018 held in NCBS, Bangalore.
- 8) Served as a volunteer and participant at the 3rd DST SERB school of chemical ecology in 2017.

Teaching/Outreach

- 1) I curated a lecture for undergraduate students in India as part of a larger [lecture series](#) on feedback across scales in biology. I specifically spoke about ecological and evolutionary feedback systems which included some basic evolutionary and ecological concepts.
- 2) I gave a popular science [talk](#) (open to the public) on colour diversity and evolution in butterflies.
- 3) Organised and participated in two editions of a scientific discussion series called ‘Sympotein’ a student driven initiative where students present topics unrelated to their thesis or area of study.

Referees

- 1) Prof. L. S. Shashidhara (lsshashidhara@ncbs.res.in), Centre Director at NCBS
- 2) Prof. Krushnamegh Kunte (krushnamegh@ncbs.res.in), PhD supervisor
- 3) Prof. Axel Brockmann (axel.ncbs@gmail.com), Thesis Committee member
- 4) Prof. Radhika Venkatesan (rv@iiserkol.ac.in), Thesis Committee member